Hasan Mohd Hussain

Data Engineer | ETL Pipelines | Python, SQL & AWS
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PROFESSIONAL SUMMARY

Data Engineer with an **MS in Computer Engineering**, building **ETL/ELT data pipelines**, **API data ingestion** from **145+ sources** and **data quality** automation in **Python** and **SQL**. Shipped a **RAG ingestion pipeline** with **30%** better retrieval and **automated validation** that cut QA time by **95%**. Experienced with **PostgreSQL**, **MongoDB**, **AWS**, **Docker** and **CI/CD**.

EDUCATION

University of Texas at Dallas — **Richardson, Texas, USA**
Master of Science in Computer Engineering — Aug 2024 - Aug 2026
GPA: 3.74/4.0 | *Relevant Coursework: NLP, Machine Learning, Computer Architecture, Embedded Systems*

University of Mumbai — **Mumbai, Maharashtra, India**
Bachelor of Engineering in Computer Engineering — Aug 2017 - Jul 2021
GPA: 7.88/10.0 | *Relevant Coursework: Data Structures, Database Management Systems*

SKILLS

Languages: Python, SQL (joins, CTEs, window functions), PL/SQL, Java, C++, Bash
Data Engineering: ETL/ELT, data pipelines, ingestion, transformation, cleansing, validation, data modeling, data warehousing, deduplication
Big Data & Orchestration: Databricks, Delta Lake, Apache Spark (PySpark), Apache Airflow, Apache Kafka, Amazon MSK, dbt, notebooks
Databases & Warehousing: Snowflake, Oracle, PostgreSQL, MySQL, SQLite, MongoDB, ChromaDB, indexing, query optimization
Cloud & DevOps: AWS (Glue, S3, Lambda, EC2, ECS Fargate, ECR), Docker, Kubernetes, CI/CD, Git, Linux, Datadog, monitoring, logging
Analytics & Quality: Pandas, NumPy, FastAPI, REST APIs, PyTest, data quality, Oracle Data Integrator (ODI), Oracle Analytics Cloud (OAC)
AI Tools: OpenAI API, Ollama, Claude Code, Claude Cowork

WORK EXPERIENCE

Mphasis — **New York, United States**
Machine Learning Engineer — May 2025 - Aug 2025, Jun 2026 - Aug 2026
- Developed an **automated QA and data validation agent** using **Python, Flask and Playwright** for a production mortgage dashboard.
- Automated **LLM-as-a-judge evaluation** across **Claude, GPT-4o, Gemini and Qwen** with **PDF/Excel reports**, cutting test time by **95%**.
- Built a **synthetic data generation pipeline** with answer keys for **5 financial document types**, producing **200 evaluation cases per run**.
- Built a **RAG ingestion and retrieval pipeline** with **ChromaDB, FastAPI and Azure OpenAI**, improving retrieval by **30%**.
- Optimized **Python backend logic** for **data-serving APIs**, improving **API response time by 20%**.

Dot9 Games — **Mumbai, India**
Game Developer Intern — Apr 2024 - Aug 2024
- Built gameplay systems in **Unity** and **C#** from design documents, shipping features that raised **user engagement by 25%**.
- Ran **POC testing** and **QA** on new game designs and optimized performance to a stable **60 FPS**, managing code with **Git**.

Shree Infotech — **Mumbai, India**
Software Engineer — Apr 2023 - Mar 2024
- Developed **Python/Flask backend services** and **REST APIs** on **Docker**, automating workflows and saving **10+ hours/week**.
- Designed **MongoDB** schemas and indexes for **fast query retrieval**, and wrote **unit tests** (PyTest) to keep core services reliable.

Nilee Games — **Mumbai, India**
Game Developer Intern — Nov 2022 - Mar 2023
- Built a **2D physics-based driving game** in **Unity/C#** within **two weeks**, contributing to **500K downloads** in its first month.
- Raised **user engagement by 30%** through gameplay tuning and **QA testing** on Android devices, using **Git** for source control.

PROJECTS

JobQuest.AI | *Python, FastAPI, SQLite, REST APIs, Scheduled ETL*
- Built an **ETL pipeline** ingesting postings from **145+ job boards** through **Greenhouse, Lever, Ashby and Workday APIs** every **12 hours**.
- **Normalized and deduplicated** records into **SQLite**, feeding **Python analytics** and **Chart.js dashboards**, backed by **59 automated tests**.

AI SQL Data Analyst Agent | *LangChain, Gemini, SQLite, SQL*
- Built a **text-to-SQL agent** that generates and executes **multi-table SQL queries** on the **Chinook relational database**.
- Returned **query results and data insights** for **natural-language analytics questions** through an interactive CLI.

RAG Document Pipeline | *LangChain, ChromaDB, Hugging Face*
- Built **PDF ingestion, text chunking and embedding pipelines** using **Hugging Face** and **ChromaDB semantic search**.
- Stored **vector embeddings** with **source metadata** for **context-grounded retrieval** of document passages.

CERTIFICATIONS

Anthropic: Claude Code in Action, Claude Code 101, Claude 101 | **Udemy:** SQL Bootcamp | **DeepLearning.AI:** AI For Everyone
IBM: Machine Learning with Python, Building AI Agents and Agentic Workflows | **Meta:** Back-End Developer Professional Certificate